ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-4 **WINTER SEMESTER, 2023-2024**

DURATION: 30 MINUTES **FULL MARKS: 15**

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. State the fundamental theorems of calculus. Applying these theorems, explain how Euler's method is used to approximate the value of a definite integral. $3+2$ (CO4) (PO1)

Solution:

The fundamental theorems of calculus are —

i. Let f be a continuous real-valued function defined on a closed interval $[a, b]$. Let F be the function defined, for all x in $[a, b]$ by

$$F(x) = \int_a^x f(t)dt$$

Then F is uniformly continuous on $[a, b]$ and differentiable on the open interval (a, b), and $F'(x) = f(x)$ for all x in (a, b) so F is an antiderivative of f.

ii. If f is Riemann integrable on $[a, b]$ then

$$\int_a^b f(x)dx = F(b) - F(a)$$

The idea is to rewrite the integral as the solution of an ODE,

$$\frac{dy}{dx} = f(x),\ y(a) = 0, \text{ using the 1st Fundamental theorem}$$

where then $y(b)$ gives us the value of the integral $\int_a^b f(x)dx$, according to the 2nd Fundamental theorem. We can use Euler's method to approximate the value of the dependent variable y at the discrete point $x = b$.

Rubric:

- 1.5 points each correct fundamental theorem.
- 2 points for the correct explanation.

2. Compare and contrast the following numerical algorithms for solving 1st order ODEs — Euler's method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method. 2 (CO3) (PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	**Euler**	**Runge-Kutta 2nd Order**	**Runge-Kutta 4th Order**
Formula	$y_{i+1} = y_i + f(x_i, y_i)h$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2)h$, $k_1 = f(x_i, y_i)$, $k_2 = f(x_i + ph, y_i + qk_1h)$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h$
Terms of the Taylor series	First 2 terms.	First 3 terms.	First 5 terms.
Order of the Taylor polynomial	1st order — which is why it's also called the Runge-Kutta 1st order method.	2nd order — which is why it's called the Runge-Kutta 2nd order method.	4th order — which is why it's also called the Runge-Kutta 4th order method.
Local Truncation Error	$O(h^2)$	$O(h^3)$	$O(h^5)$
Global Truncation Error	$O(h)$	$O(h^2)$	$O(h^4)$
Variant Approaches	None.	Has 3 different versions – Heun's method, Midpoint method, and Ralston's method	Has 2 different versions – Runge's approach and Kutta's approach.

Rubric:

- 1 point for each valid comparison.

3. Observe the lightly shaded solid object portrayed in Figure 1.

It is a curved wedge cut from a circular cylinder of radius 3 by two planes. One plane is perpendicular to the axis of the cylinder. The second plane crosses the first plane at a $45°$ angle at the center of the cylinder. The wedge's base is the semicircle with $x \geq 0$ that is cut from the circle $x^2 + y^2 = 9$ by the $45°$ plane when it intersects the y-axis. When this solid object is cut by a plane perpendicular to the x-axis and the xy-plane, we obtain a cross-section at x which is a rectangle of height x whose width extends across the semicircular base (as indicated by the darkly shaded cross-sectional area).

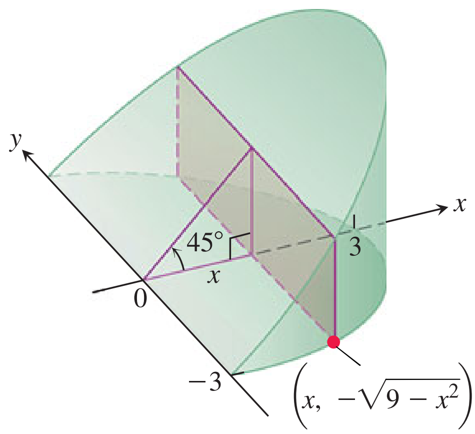

Figure 1: A solid geometric object.

a) Formulate an integrand function $f(x)$ whose integral value within the limits $[0, 3]$ will yield the volume of the solid object. 1.5 (CO1) (PO1)

Solution:
There are 3 key observations here—

- The cross-sections of the object are rectangles.
- The equation of the semi-circle is $y = \pm\sqrt{9 - x^2}$, which makes the lengths of the rectangles to be $2\sqrt{9 - x^2}$.
- The wedge has been cut at an angle of $45°$, which means the right-angled triangle formed by the wedge's base, roof, and cross-section is an isosceles right-angled triangle. This implies that the breadths of the rectangles are increasing according to the increasing nature of the $y = x$ line.

We know, that the area of a rectangle having a length of l and a breadth of w is lw.
Here, for a particular value of x, the length of an infinitesimally thin rectangle cross-sectional slice is $2\sqrt{9 - x^2}$ and the breadth is x. So, the area of that rectangular slice will be $2x\sqrt{9 - x^2}$. All of these infinitesimally thin rectangular slices within the limits $[0, 3]$ together constitute the entire volume of the solid object, *ergo* the integrand is,

$$f(x) = 2x\sqrt{9 - x^2}$$

Rubric:

- 1.5 points for a correct integrand function.

b) Approximate the volume of the solid object using Mixed Simpson's rule *i.e.* Simpson's 1/3 rule (with $n_1 = 4$) and Simpson's 3/8 rule (with $n_2 = 3$). 4.5 (CO2) (PO1)

Solution:

Here, $h = \dfrac{b-a}{n} = \dfrac{b-a}{n_1+n_2} = \dfrac{3-0}{7} \approx 0.42857$. Let's compute the functional values at the right-hand side of each of these $n = 7$ segments.

$$
\begin{aligned}
x_0 &= a = 0; & f(x_0) &= 0\\
x_1 &= a + h = 0 + 0.42857; & f(x_1) &= 2.545\\
x_2 &= x_1 + h = 0.857; & f(x_2) &= 4.928\\
x_3 &= x_2 + h = 1.285; & f(x_3) &= 6.969\\
x_4 &= x_3 + h = 1.714; & f(x_4) &= 8.440\\
x_5 &= x_4 + h = 2.142; & f(x_5) &= 8.998\\
x_6 &= x_5 + h = 2.571; & f(x_6) &= 7.946\\
x_7 &= x_6 + h = 3; & f(x_7) &= 0
\end{aligned}
$$

The multiple-segment equation for Simpson's 1/3 rule is

$$I_1 = \int_a^b f(x)dx \approx \frac{b-a}{3n_1}\left[f(x_0) + 4\sum_{\substack{i=1,\\ i=\text{odd}}}^{n_1-1} f(x_i) + 2\sum_{\substack{i=2,\\ i=\text{even}}}^{n_1-2} f(x_i) + f(x_{n_1})\right]$$

where, $a = 0$ and $b = x_4 = 1.714$. Plugging in the respective values, we get,

$$
\begin{aligned}
I_1 &\approx \frac{1.714}{3\times 4}[0 + 4\times(2.545 + 6.969) + 2\times(4.928) + 8.440]\\
&= 8.048
\end{aligned}
$$

The multiple-segment equation for Simpson's 3/8 rule is

$$I_2 = \int_a^b f(x)dx \approx \frac{3\times(b-a)}{8n_2}\left[f(x_{n_1}) + 3\sum_{\substack{i=n_1+1,\\ i\%3\neq 0}}^{n_1+n_2-1} f(x_i) + 2\sum_{\substack{i=n_1+3,\\ i\%3=0}}^{n_1+n_2-3} f(x_i) + f(x_{n_1+n_2})\right]$$

where, $a = x_4 = 1.714$ and $b = x_7 = 3$. Plugging in the respective values, we get,

$$
\begin{aligned}
I_2 &\approx \frac{3\times 1.285}{8\times 3}[8.440 + 3\times(8.998) + 2\times(7.946) + 0]\\
&= 8.244
\end{aligned}
$$

So the volume of the solid object is $I = \displaystyle\int_0^3 2x\sqrt{9-x^2}dx \approx I_1 + I_2 = 8.048 + 8.244 = 16.293$.

Rubric:

- 2 points for a correct I_1 value.
- 2 points for a correct I_2 value.
- 0.5 point for an approximately correct answer.

4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a math meme related to this course. 2 (CO69) (PO420)
Warning: Cringe and lame memes will result in negative marking.

Solution:

Figure 2: Real.

Rubric:

- 2 points for a dank meme.